

Question 3 of 5

$$\begin{aligned} a) \quad E(X) &= 3 \\ E(X^2) &= 13 \quad \Rightarrow \text{Var}(X) = E(X^2) - [E(X)]^2 \\ &= 13 - 9 = 4 \end{aligned}$$

$$\begin{aligned} &P(-2 < X < 8) \\ &= P(-2-3 < X-3 < 8-3) \\ &= P(-5 < X-E(X) < 5) \\ &= P(|X-E(X)| < 5) \\ &= 1 - P(|X-E(X)| \geq 5) \end{aligned} \quad (8)$$

$$\begin{aligned} P(|X-E(X)| \geq 5) &\leq \frac{\text{Var}(X)}{5^2} = \frac{4}{25} \\ \Rightarrow P(-2 < X < 8) &= 1 - P(|X-E(X)| \geq 5) \geq 1 - \frac{4}{25} \\ &= \frac{21}{25} \end{aligned}$$

$$\begin{aligned} b) \quad E(X) &= 0, \text{Var}(X) = 1 \\ P(|X| \geq k) &\leq 0.01 = \frac{1}{100} \end{aligned} \quad (7)$$

Using Chebyshev's inequality

$$P(|X| \geq k) = P(|X-E(X)| \geq k) \leq \frac{\text{Var}(X)}{k^2} = \frac{1}{k^2}$$

$$\Rightarrow \frac{1}{100} = \frac{1}{k^2} \Rightarrow k = 10$$

$$\begin{aligned}
 e) \quad & P(X_i > x) \\
 &= P(X_1 > x, X_2 > x, \dots, X_n > x) \\
 &= \prod_{i=1}^n P(X_i > x) \\
 &= (1 - F(x))^n
 \end{aligned}$$

(3)

$$P(X_i \leq x) = 1 - (1 - F(x))^n$$

$$\begin{aligned}
 \text{pdf of } X_i, f_{X_i}(x) &= \frac{d}{dx} P(X_i \leq x) \\
 &= n(1 - F(x))^{n-1} f(x)
 \end{aligned}$$

(2).